Project Documentation: Sacramento Explorer

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Course: Web Site Development
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GitHub Repository Link: https://github.com/delossantosedelia/sacramento-explorer
Live Website Link: https://github.com/delossantosedelia/sacramento-explorer

1. Project Title and Overview

- **Project Title:** Sacramento Explorer
- **Overview:** This project is an interactive, responsive single-page web application designed to help tourists and locals discover key attractions in Sacramento. The website features a clean layout, interactive filtering, a detailed modal window, and a dark theme toggle. No major changes were made from the initial proposal, as the pre-planned frontend mechanics were successfully implemented.

2. Target Audience

- **Audience Profile:** Tourists visiting Northern California, new residents exploring local communities, and weekend commuters looking for curated activities in Sacramento.
- **Addressing Their Needs:** The interface provides a lightweight, distraction-free environment that functions perfectly on mobile phones while walking around the city. It allows users to quickly sort locations by category (food, parks, or culture) without waiting for a webpage to reload and includes a dark theme option for comfortable night-time reading.

3. Core Features & Technology Matrix

- **Interactive Dark Theme Toggle:** Powered by **HTML5** (semantic button layout), **CSS3** (custom variables mapping colors under body dark theme and layout transitions), as well as **JavaScript** (a click listener toggling classes on the body and rewriting button labels).
- **Responsive Destination Grid:** Managed via **HTML5** structure and **CSS3** Grid Layout using repeat (auto-fit, minmax(300px, 1fr)), ensuring the interface automatically scales down layout grids perfectly on small devices.
- **Dynamic Category Filter:** Engineered with **HTML5** target attributes (data-filtered) **CSS3** classes for active button styling states, and **JavaScript** logic that

monitors clicked attributes to switch card displays between block and none instantly.

- **Details Modal Canvas Popup:** Handled with a fixed **HTML5** structural window block, positioned explicitly using **CSS3** overlay elements, and hydrated with dynamic text strings drawn from a **JavaScript** lookup dictionary (venueDetails) when a user selects a card button.

4. Design and UX Considerations

- **Visual Interface:** Built upon a clean grid layout utilizing accessible grays (#f8f9fa) and whites, accented with a strong foliage green (#1e7e34) to represent Sacramento’s identity as the "City of Trees."
- **User Experience (UX) Optimizations:** Implemented position: sticky on the navigation bar to keep controls permanently accessible. Applied object-fit: cover on images to guarantee unified container spacing across various viewports, preventing layout shifts.

5. Challenges and Solutions

- **The Challenge:** During development, attraction images displayed unevenly inside the CSS grid, distorting proportions and squishing layouts on mobile devices. Furthermore, an image asset failed to load properly upon deploying to GitHub due to case-sensitivity restrictions.
- **The Solution:** The CSS selector was targeted precisely to .card img to force explicit aspect rules (height: 200px) while applying object-fit: cover to preserve visual ratios. The file naming discrepancies were corrected inside the HTML source to match GitHub's directory architecture exactly.

6. Lessons Learned

- Gained an advanced understanding of decoupling website styles from structural code blocks by utilizing native CSS custom variables.
- Discovered the power of responsive fluid CSS layout grids instead of hardcoding heavy, repetitive layout breakpoints for varying phone display sizes.
- Learn the importance of strict file structuring and naming rules when deploying frontend directories to live production environments.